Lab 1: Use-Case Flow Specification

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Problem Statement #41: OpenAPI Mock Server Generator

Use Case ID / Name	UC-02: Generate Mock Endpoints
Primary Actor	API Developer
Secondary Actor	QA Engineer (consumes the generated endpoints)
Trigger	The API Developer selects an uploaded OpenAPI spec and clicks "Generate Mock Endpoints".
Preconditions	1. The tool is running and the API Developer is on the dashboard. 2. An OpenAPI 3.0 YAML/JSON specification has been uploaded and is stored in the system. 3. The mock server port is free.
Postconditions (success)	1. Mock endpoints for every path/operation in the spec are registered and reachable. 2. Each endpoint returns randomized, schema-compliant JSON and uses its configured (or default 0 ms) latency. 3. The generated endpoint list is shown to the user and the event is logged.
Postconditions (failure)	No endpoints are created or changed; the previous mock configuration (if any) stays active; the error is shown and logged.
Includes	UC-03: Parse & Validate OpenAPI Spec

Main Success Scenario

1. The API Developer selects an uploaded specification and chooses **Generate Mock Endpoints**.
2. The system loads the specification from the Spec Store.
3. The system parses and validates the specification (*include* UC-03); validation passes.
4. The system extracts every path, HTTP method, parameter set and response schema.
5. For each operation, the system creates a dynamic route in the Mock Server Engine and attaches the request validator and the response generator.
6. The system applies the default latency (0 ms) or any latency already configured for the endpoint.
7. The system starts (or hot-reloads) the mock server and confirms that all routes are active.
8. The system displays the list of generated endpoints (method, URL, status codes) and logs the generation event.
9. The API Developer copies the base URL; the use case ends successfully.

Alternate Flow A1: Specification is invalid (branches from step 3)

3a. The parser finds a syntax error, unsupported OpenAPI version (not 3.0), unresolved `$ref`, or a remote `$ref` that is not allowed.
3b. The system stops processing, creates no endpoints and leaves any existing mock configuration unchanged.
3c. The system displays each error with its line number / JSON path (for example, `#/paths/~1users/get/responses`) and logs the failure.
3d. The API Developer corrects the specification and re-uploads it (UC-01), then the flow resumes at step 1. If the developer cancels, the use case ends in the failure postcondition.

Related Requirements

FR-001, FR-004, NFR-002 (safe spec handling)